ECEN 153 HW 2

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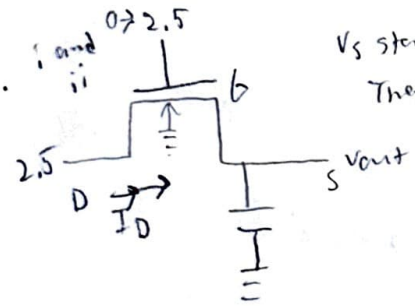
1. $K_n' = 115\,\mu A/V^2$ $V_{TN0} = 0.43V$ $\lambda_n = 0$ $\gamma = 0$ $V_{dsat} = .63V$ $V_{TN} = .6V$ $W/L = 20$

$K_p' = 30\,\mu A/V^2$ $V_{TP0} = -.4V$ $\lambda_p = 0$ $\gamma = 0$ $V_{dsat} = -1V$ $V_{TP} = -.55V$

$V_{DD} = 2.5$

a. i and ii



V_S starts at 0 but changes.
There is body effect

at start
$V_S = V_B$ source set
$V_{DS} = 2.5$
$V_{GT} = 2.5 - .43 = 2.07$
vs
$V_{dsat} = .63$
start

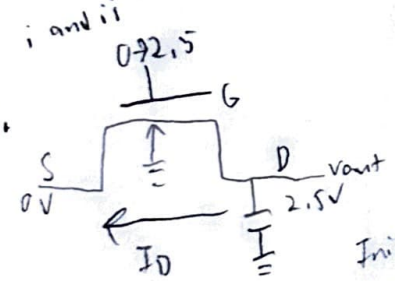
$V_{DS} = .6$
end

$V_{GT} = 2.5 - .6 = 1.9$

iii. $V_{out} = 1.9V$
iv. velocity sat
v. cutoff
vi. $\infty\,\Omega$

$I_{DS} = K_n'(\frac{W}{L})(V_{GS} - V_T)$

b. i and ii

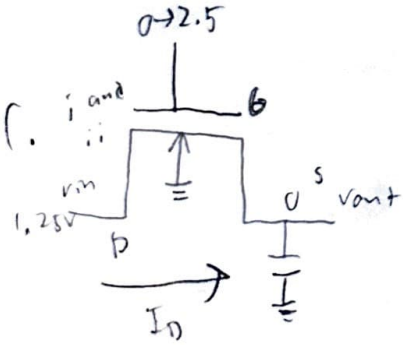


Init
$V_{ds} = 2.5$
$V_{GT} = 2.5 - .43 = 2.07$
$V_{dsat} = .63$

final
$V_{ds} = 0$
$V_{GT} = 2.07$
$V_{dsat} = .63$

iii. $V_{out} = 0V$
iv. velocity sat
v. linear
vi. $r_D = 210.04\,\Omega$

c. i and ii



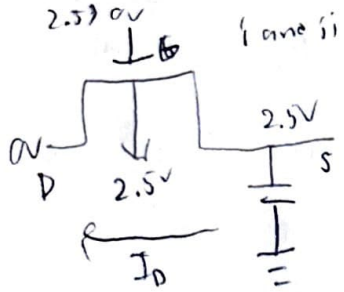
Init
$V_{ds} = 1.25$
$V_{GT} = 2.07$
$V_{dsat} = .63$

final
$V_{DS} = 0$
$V_{GT} = 2.5 - 1.25 - .6 = 1.65$
$V_{dsat} = .63$

$I_D = 1.311\,mA$

iii $V_{out} = 1.25V$
iv. velocity sat
v. linear
vi. $r_D = 668.90\,\Omega$

d.

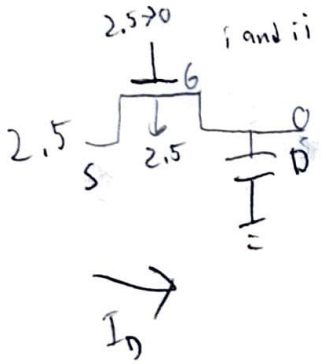


i and ii

init	final	
$V_{GT} = -1.95$	$V_{GT} = 0 + .55 - 0$	i V, velo sat
$V_{DS} = 2.5$	$V_{DS} = -.4$	V, cutoff
$V_{dsat} = -1$	$V_{dsat} = -1$	vi, $R_d = \infty\Omega$

iii. .55V

e.

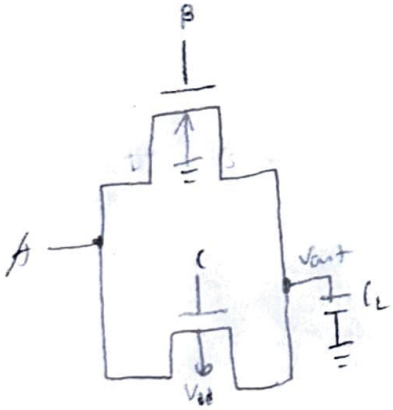


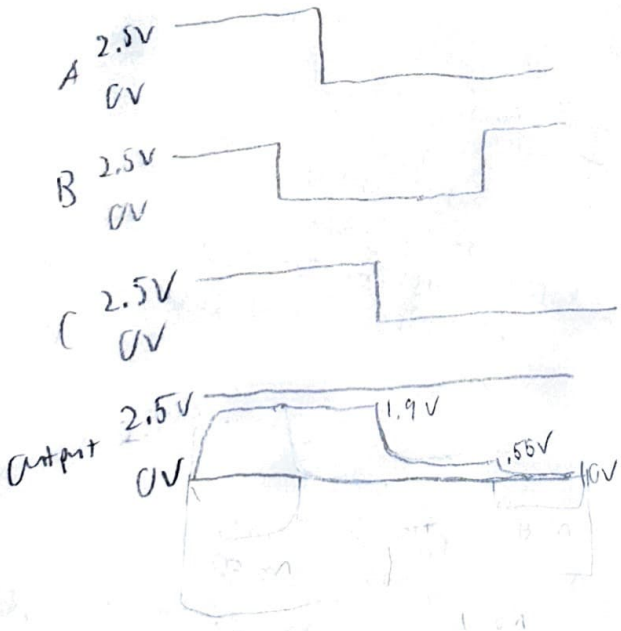
i and ii

init	final	
$V_{GT} = +.4$	$V_{GT} = 0 - 2.5 -$	iii. 2.5V
$-2.5 = -2.1$	$.4 = -2.1$	iv. velo sat
$V_{DS} = -2.5$	$V_{DS} = 0$	vi $R_d = 293.65\Omega$
$V_{dsat} = -1$	$V_{dsat} = -1$	

V, linear

2.





3.

fig3

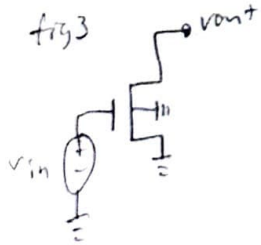


fig 3a

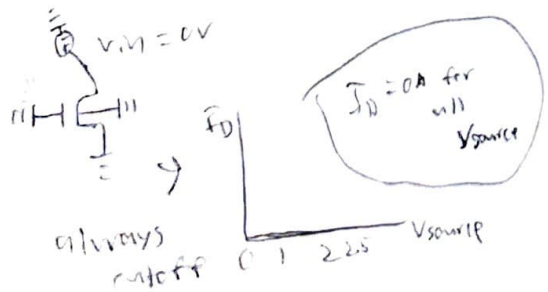
fig 3b

$I_d = k' \frac{W}{L}(V_{GT}V_{min} - \frac{V_{min}^2}{2})$

$V_{TH0} = .43$

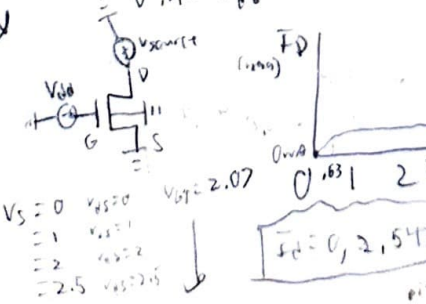
$V_{dsatn} = .63V$

9.



no body effect

y



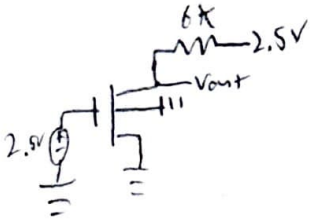
0 → .63: linear

.63 → 2.5: velo sat

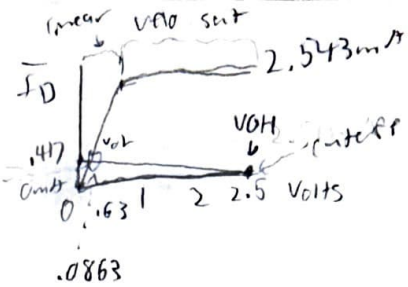
$I_d = 0, 2.543mA, 2.543mA, 2.543mA$

cutoff: Vsource = .63

b.



$\frac{2.5}{6k} = .417mA$



op point:
$(.0863, 4.023\times10^{-4})$

and r,

$y = -\frac{1}{6k}x + .417\times10^{-3}$

$V_{OL} = 86.3mV$

$V_{OH} = 2.5V$

($V_{in}=0$, so cutoff mode and no I_D. Thus $V_{out} = 2.5V$)

in linear

$\frac{2.5 - V_L}{6k} = 115\times10^{-6}\times20\left(V_L(2.07) - \frac{V_L^2}{2}\right)$

$2.5 - V_L = 13.8\left(2.07V_L - \frac{V_L^2}{2}\right)$

$2.5 - V_L = 28.566V_L - 6.9V_L^2$

$6.9V_L^2 - 29.566V_L + 2.5 = 0$

$\frac{29.566 \pm \sqrt{29.566^2 - (4)(6.9)(2.5)}}{13.8} = \frac{29.566 \pm 28.375}{13.8} = .0863V$

other answer is too big